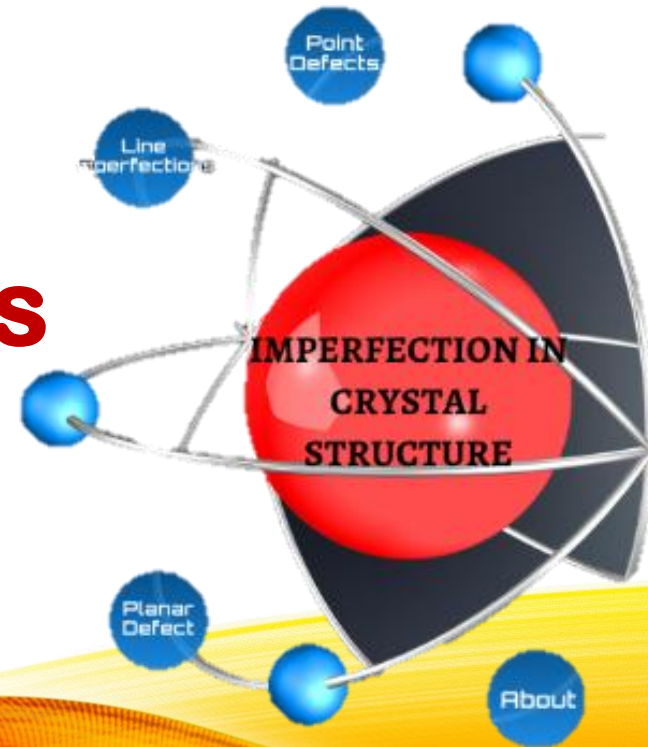


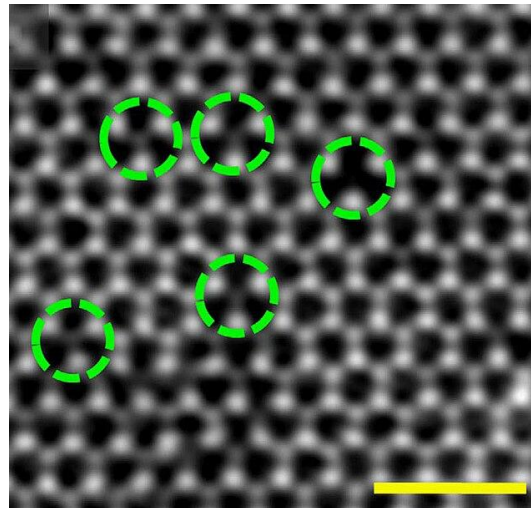
316-221: Metallurgy

Chapter 3: Imperfection in solids



INTRODUCTION ²

- The **properties** of materials are profoundly **influenced** by the presence of **imperfections**.
- It is important to have knowledge about the **types of imperfections** that exist and the roles they play in **affecting** the behavior of materials.



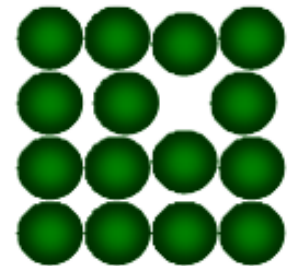
Point defects

Structural defects exist in real materials independently of chemical impurities. Imperfections associated with the crystalline point lattice are called point defect.

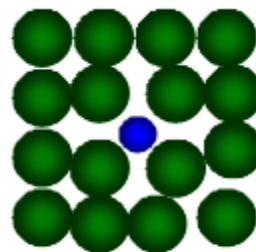
Two common types of point defect:

1) Vacancy

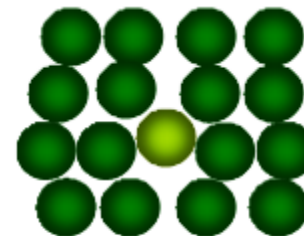
2) Self-interstitial



Vacancy



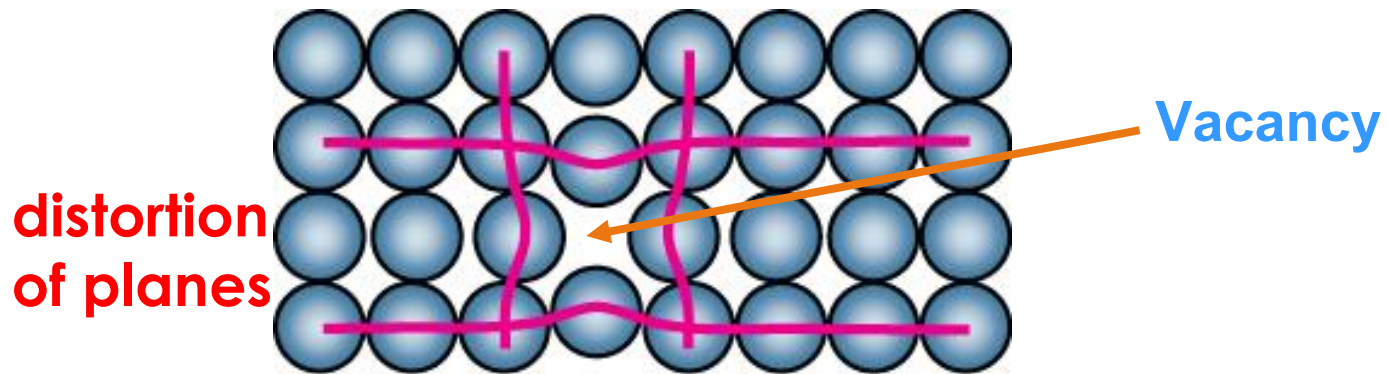
Interstitial impurity



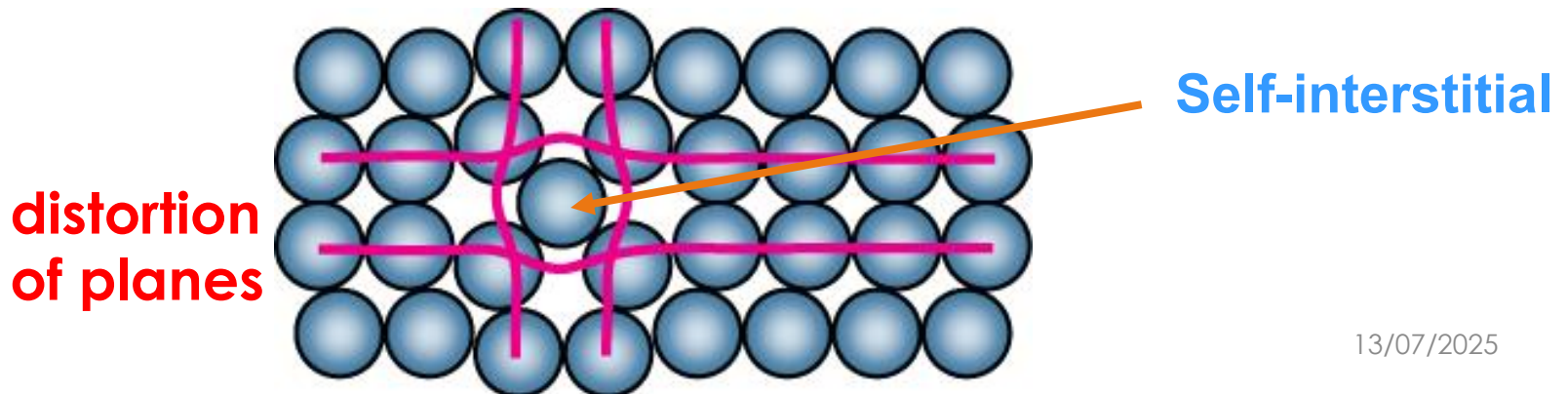
Self-interstitial

Point defects

- **Vacancies:**
 - vacant atomic sites in the crystal structure.

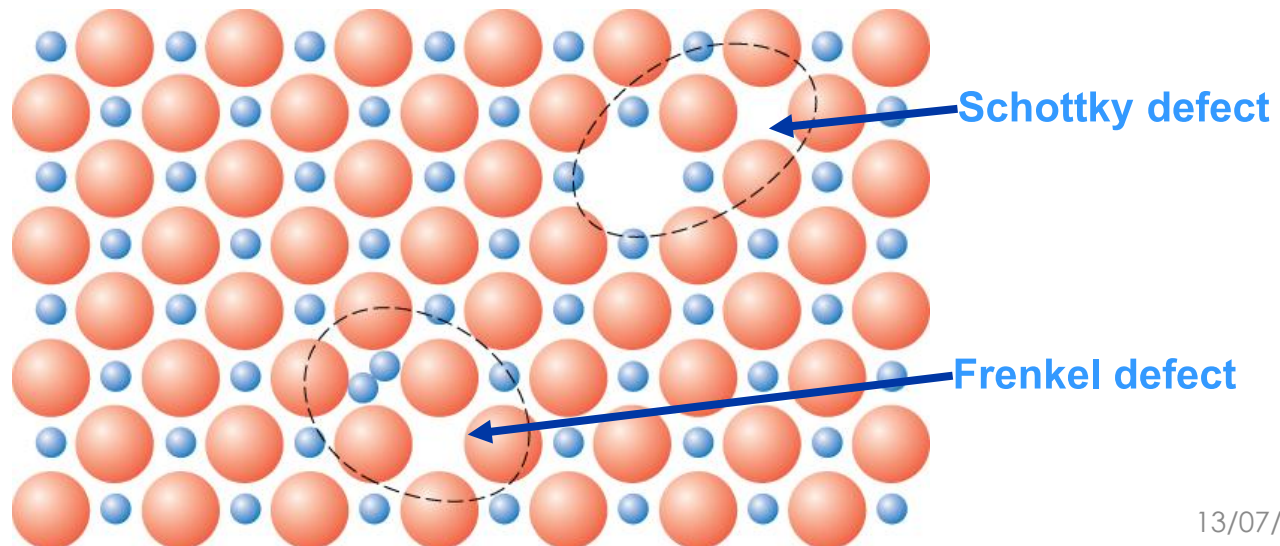


- **Self-interstitials:**
 - "extra" atoms positioned between atomic sites.



Point defect: Frenkel and Schottky

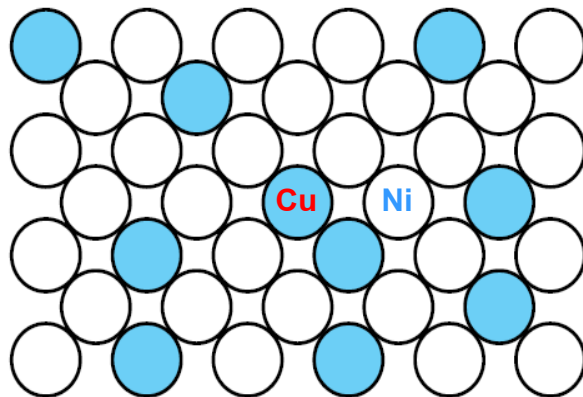
- **Frenkel Defect:** To maintain the charge neutrality, a cation vacancy-interstitial pair occur together. The cation leaves its normal position and moves to the interstitial site.
- **Schottky Defect:** To maintain the charge neutrality, remove 1 cation and 1 anion; this creates 2 vacancies.



Point defects in alloys

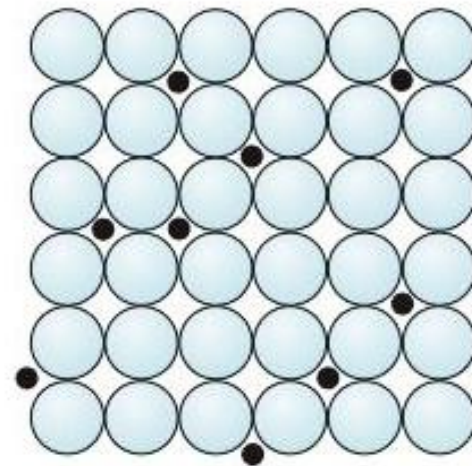
What are the outcomes if impurity (**B**) is added to host (**A**)?

- **Solid solution** of **B** in **A** (random distribution of point defects)



Substitutional
solid solution (e.g., **Cu** in **Ni**)

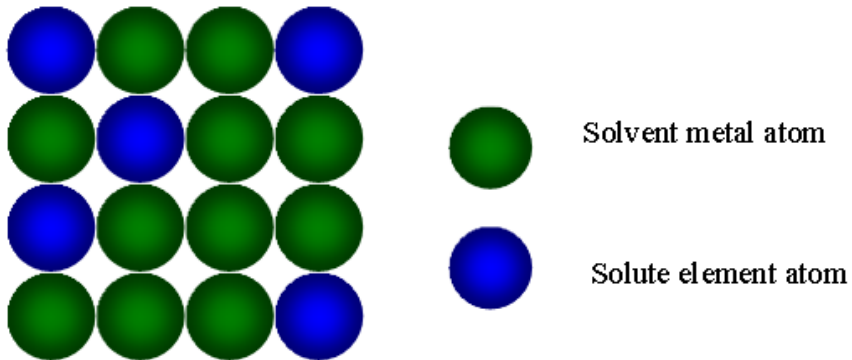
OR



Interstitial
solid solution (e.g., **C** in **Fe**)

Impurities defects: Substitutional or interstitial

Substitutional : impurity atoms type A substitutional atoms type B in structure



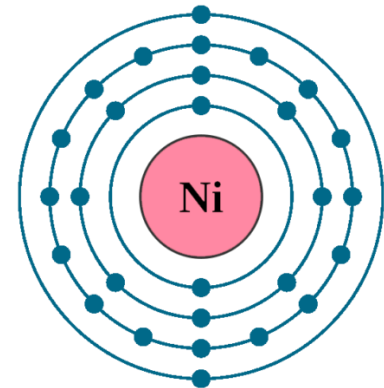
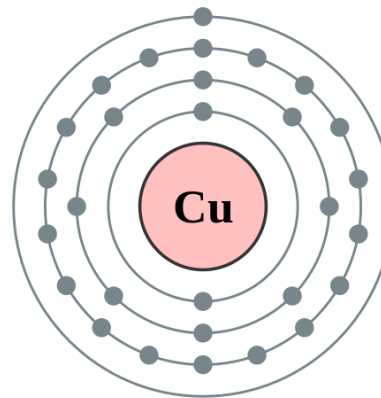
Basic conditions for an element to dissolve in a metal, forming a substitutional solid solution.

1. The **atomic radius** of the solute and solvent atoms must differ by **no more than 15%** .
2. The solute and solvent should have **similar electronegativities**.
3. The **same crystal structure**.
4. Maximum solubility occurs when the solvent and solute have the **same valence electron**.

Impurities defects: Substitutional or interstitial

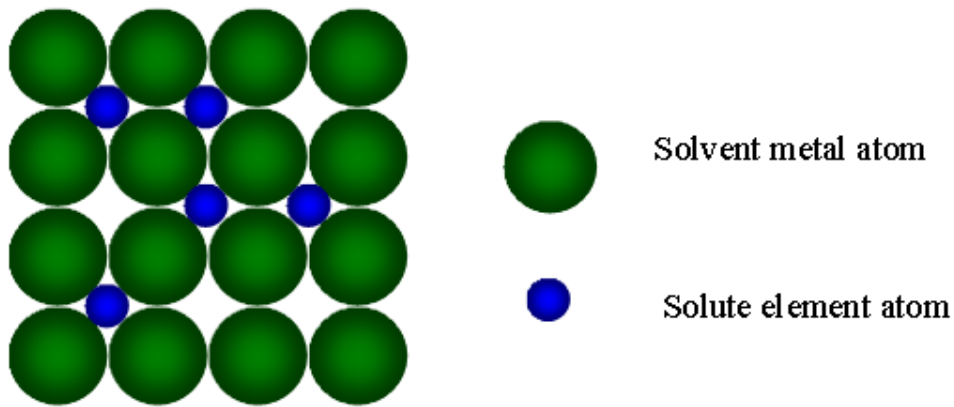
□ **Cu** and **Ni** have the same complete solubility in one another at all properties:

- **Atomic Size Factor ($\pm 15\%$)**
Atomic radii Cu = 0.128 nm, Ni = 0.125 nm
- **Crystal Structure (same)**
Both Cu and Ni have FCC crystal structure
- **Electronegativity (low)**
Cu = 1.9, Ni = 1.8
- **Valence electron**
1 for Cu, and 2 for Ni

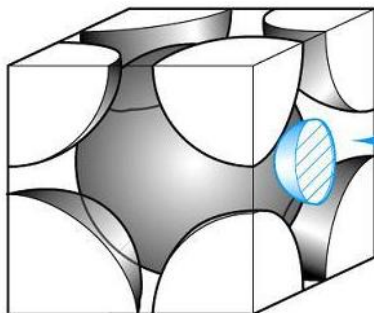


Impurities defects: Substitutional or interstitial

Interstitial : atoms type A fill the voids or interstices among the atoms type B



Interstitial solid solution of C in α -iron, The carbon atom is small enough to fit with some strain in the interstice among adjacent Fe atoms in this structure of importance to the steel industry.



C atom dissolved interstitially at a $\frac{1}{2} 0 \frac{1}{2}$ -type position in the bcc structure of α -Fe

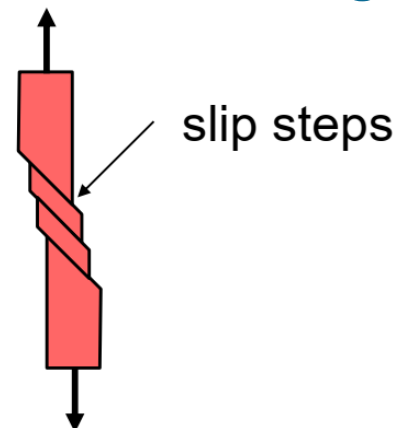
Line defects or dislocations

- Line defects are associated primarily with **mechanical deformation**. Slip between crystal planes result when dislocations move, produce **plastic deformation**.
- Line defects are also known as **dislocation**.
- Types of dislocations: **edge, screw, mixed dislocation**

■ Before deformation

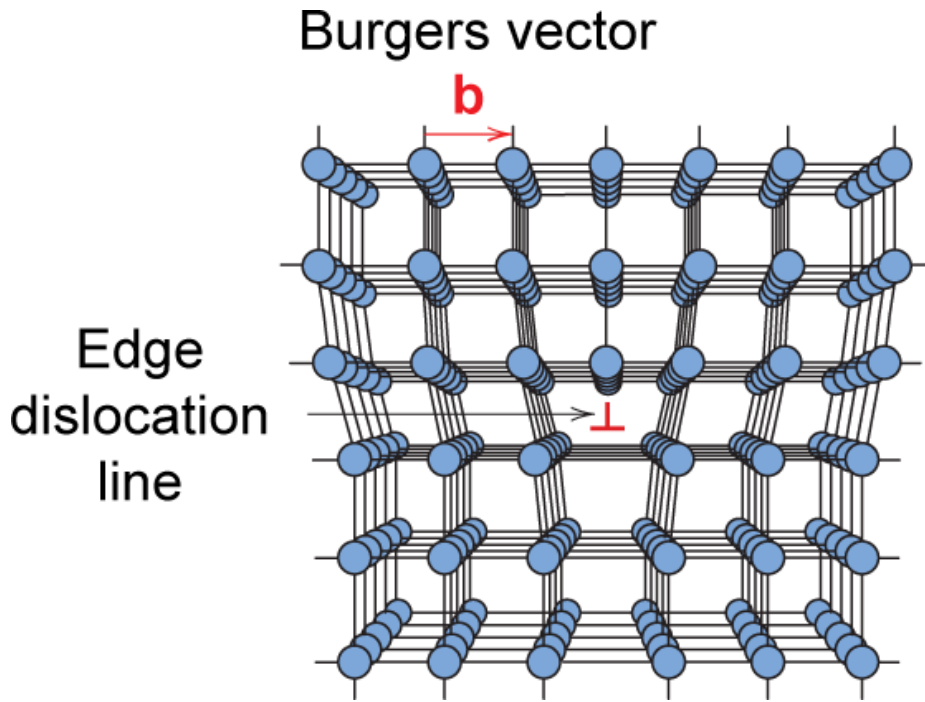


■ after tensile elongation



Line defects or dislocations

Edge dislocation:



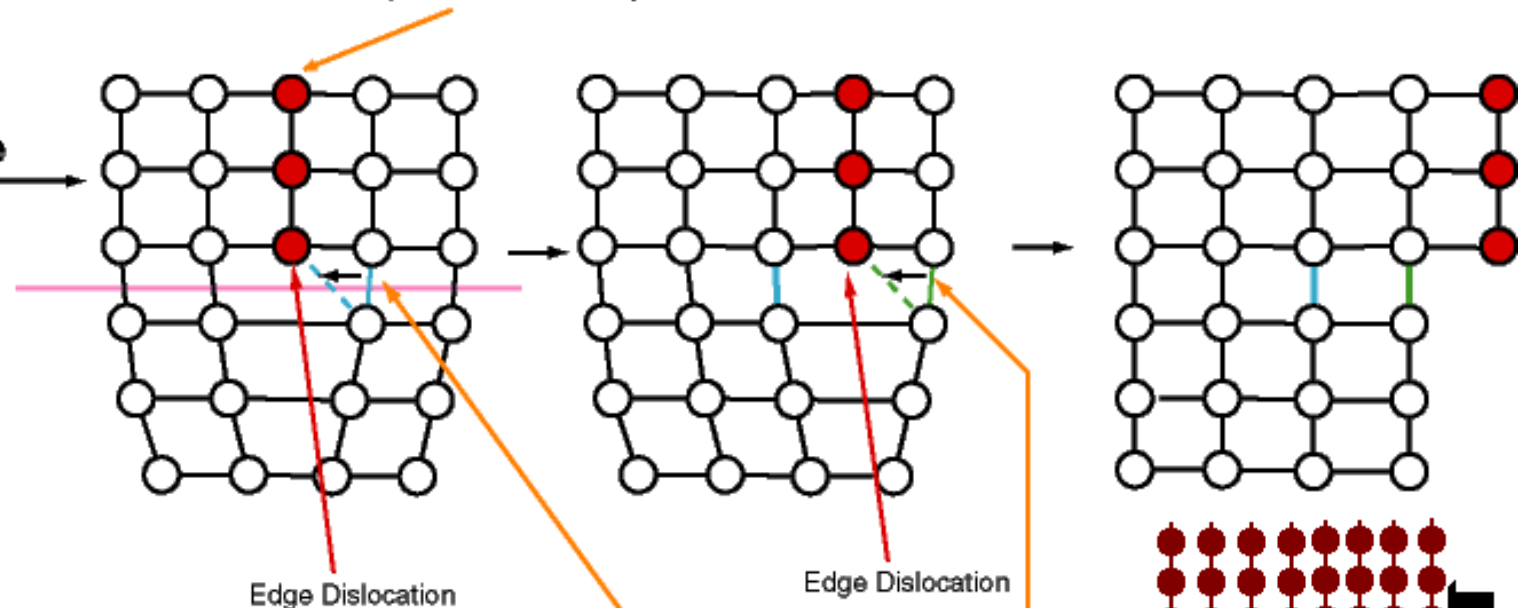
- extra half-plane of atoms inserted in a crystal structure; the edge of the plane terminates within the crystal.
- Around the dislocation line there is some localized distortion.
- $\mathbf{b}$ perpendicular ($\perp$) to dislocation line.

Line defects or dislocations

The movement of edge dislocation through the crystal

Extra plane of atoms in crystal

Force



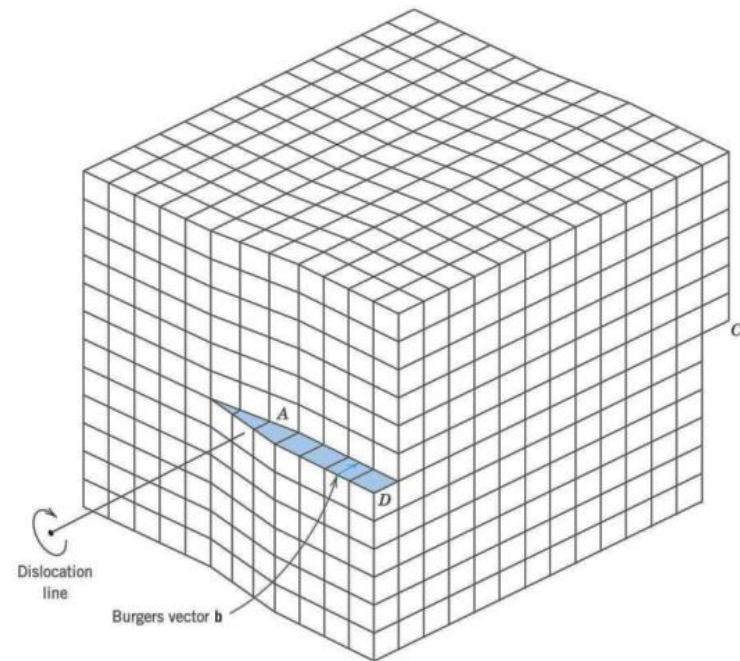
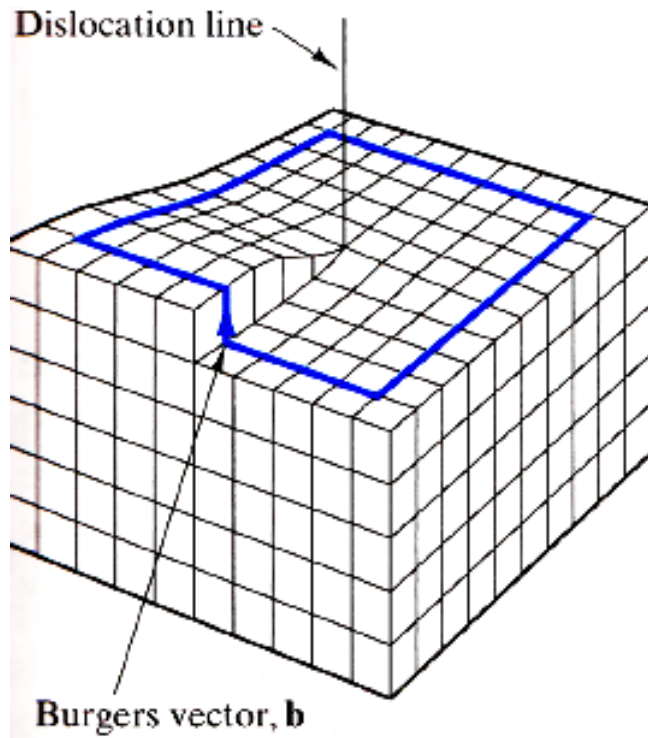
Edge Dislocation

Edge Dislocation

This row of bonds will break and reattach itself to a different row of atoms. It is much easier for only one row of bonds to break and reform than for an entire plane of bonds (i.e. the bonds intersecting the pink line) to do so.

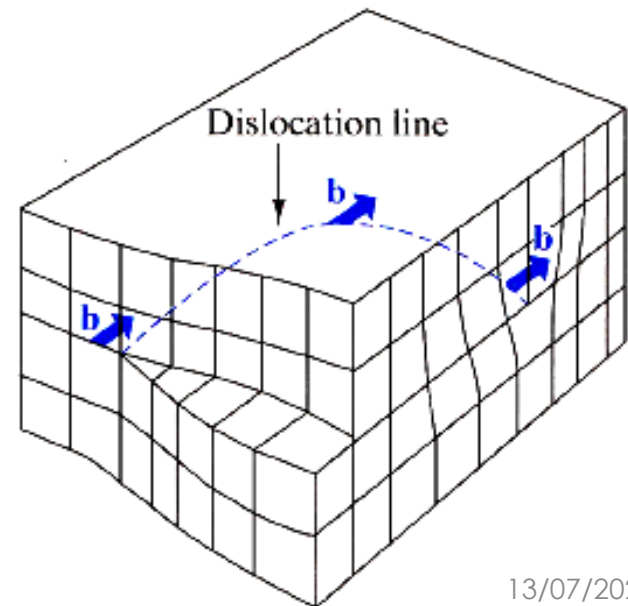
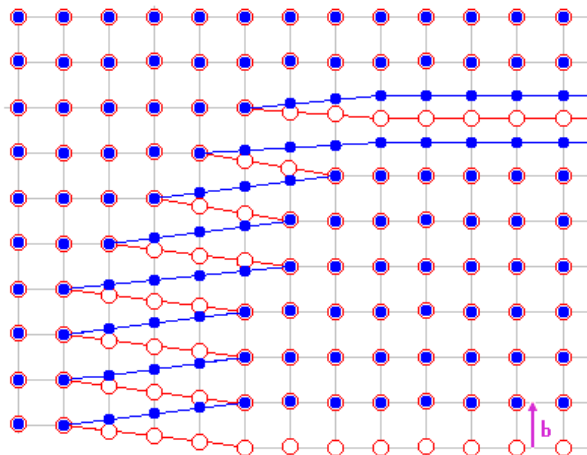
Line defects or dislocations

Screw dislocation: Burgers vector (**b**) is parallel (**||**) to the dislocation line.

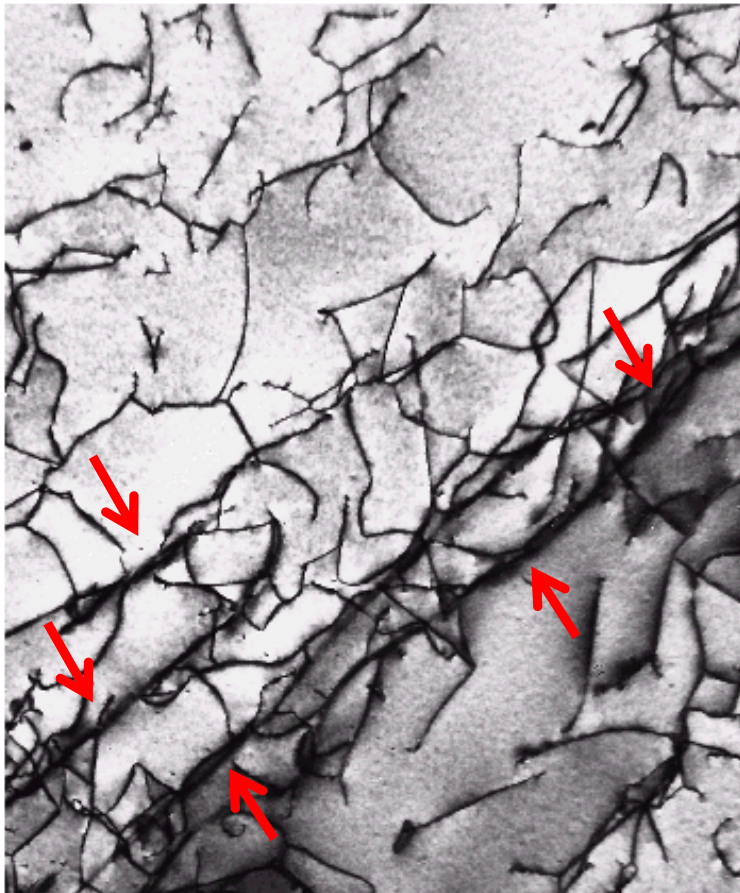


Line defects or dislocations

- **Mixed dislocation** has both **edge** and **screw** character with a single Burgers vector consistent with the pure edge and pure screw region.
- Most linear defects in actual materials will be **mixed**.



Dislocations during cold working

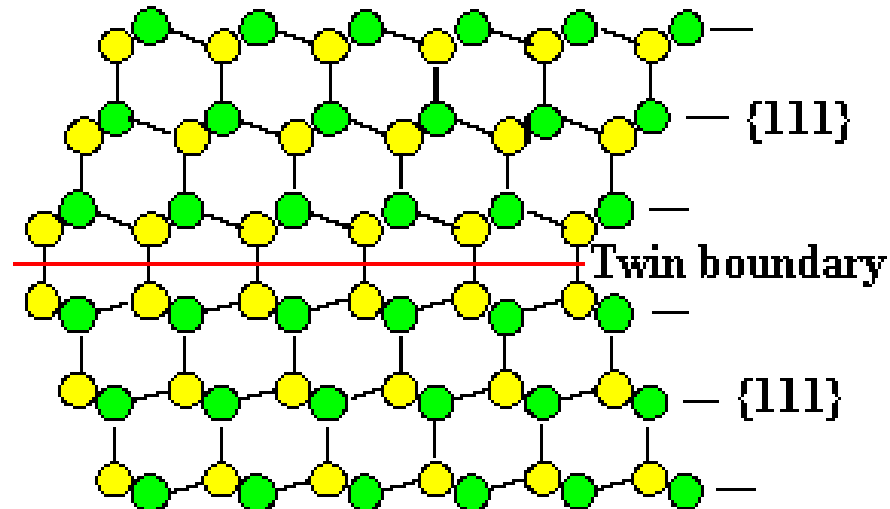


TEM image

- Ti alloy after cold working.
- Dislocations entangle with one another during **cold work**.
- Dislocation movement becomes more difficult.
- Dislocations are visible in electron micrographs.

Planar defects

A shear force that causes atomic displacements such that the atoms on one side of a plane (**twin boundary**) mirror the atoms on the other side. A reflection of atom positions across the **twin plane**.

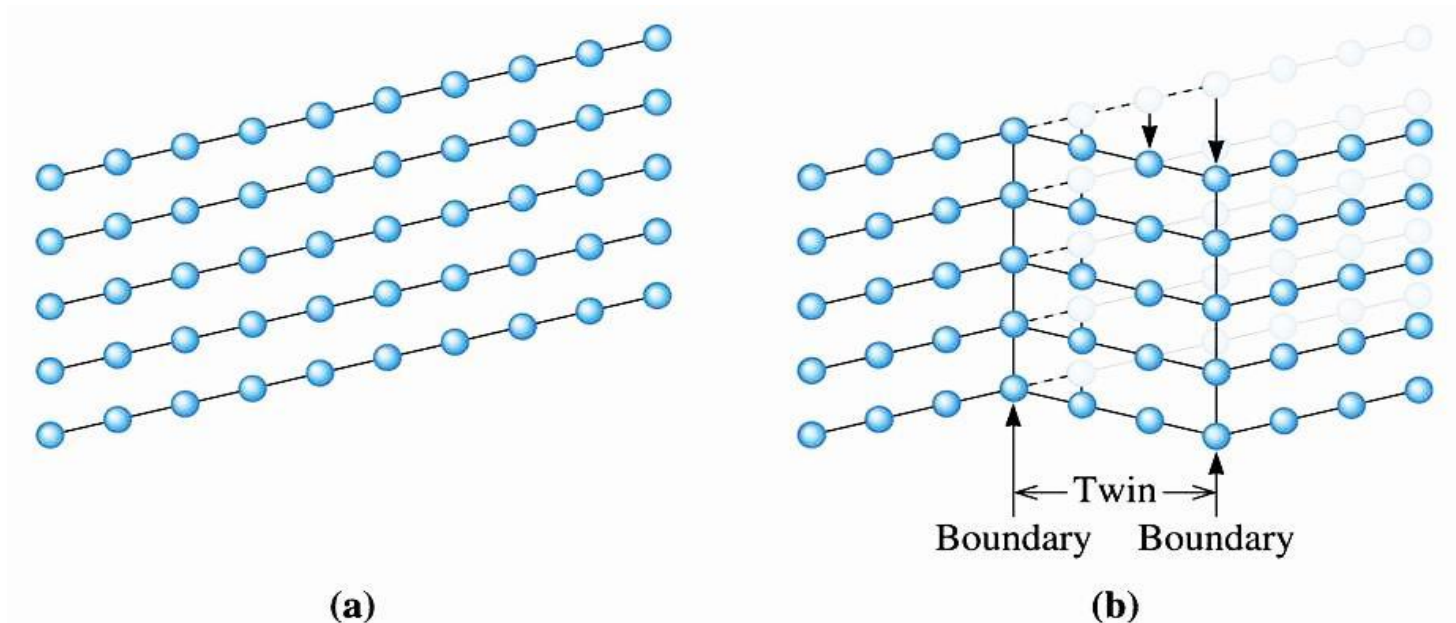


Planar defects

■ Twin boundary:

Applied stress to a **perfect crystal (a)** may cause a displacement of the atoms, **(b)** causing the formation of a twin.

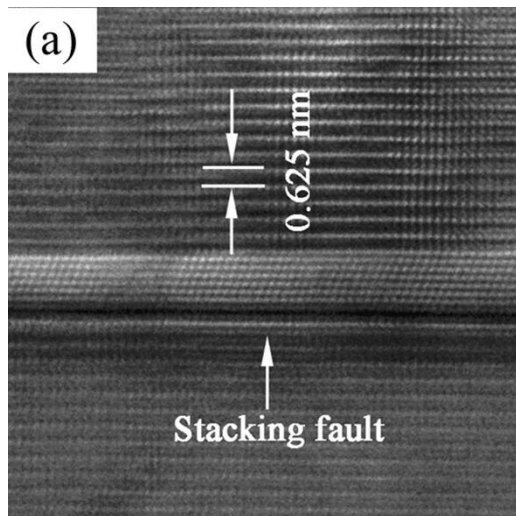
Note that the crystal has deformed as a result of twinning.



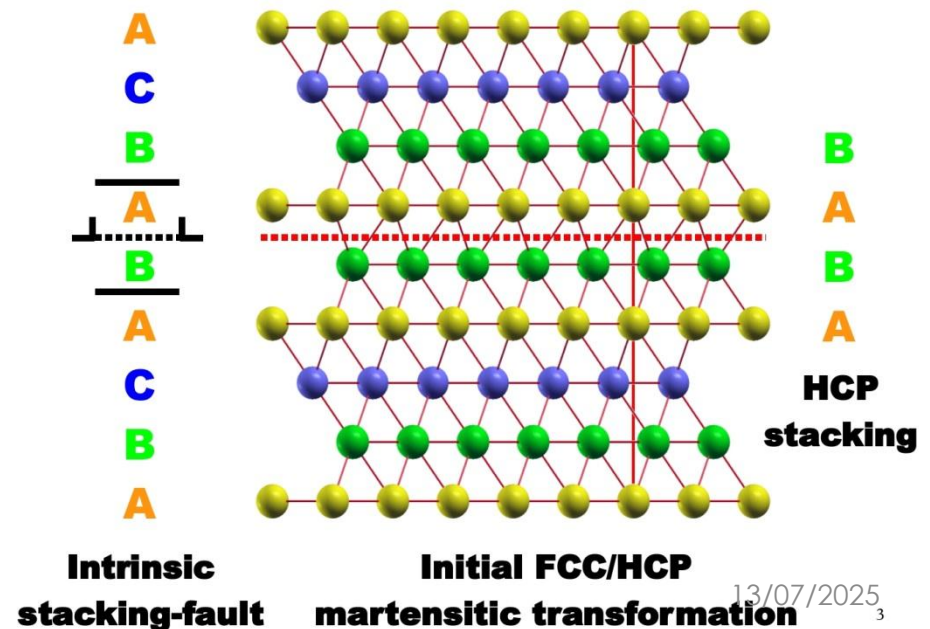
Planar defects

■ Stacking faults

- The **stacking-fault** is materials property that affects the plastic deformation.
- The **perfect FCC stacking ...ABCABCABC...** is disturbed by the removal of a lattice plane C.
- An **intrinsic stacking-fault** is created with the new stacking **...ABCAB|ABCA...**



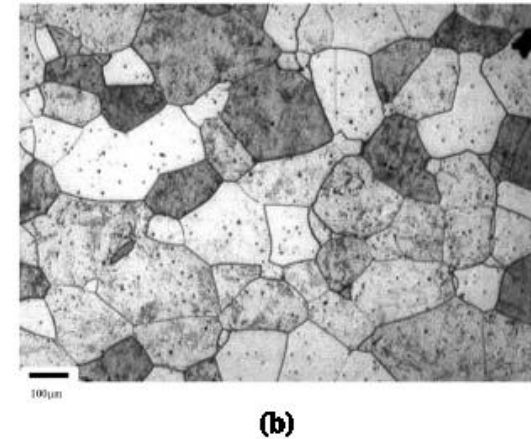
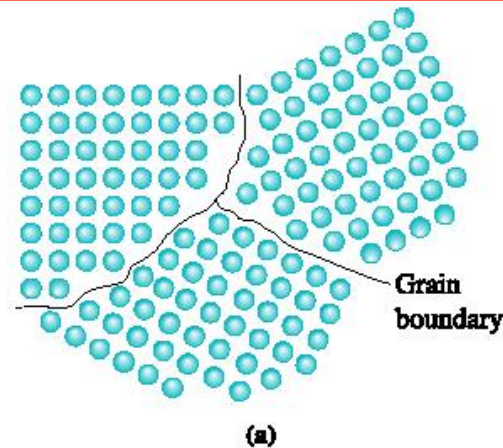
Crystal defects in ZnS nanocrystals
(a) stacking fault in a nanobelt



Planar defects

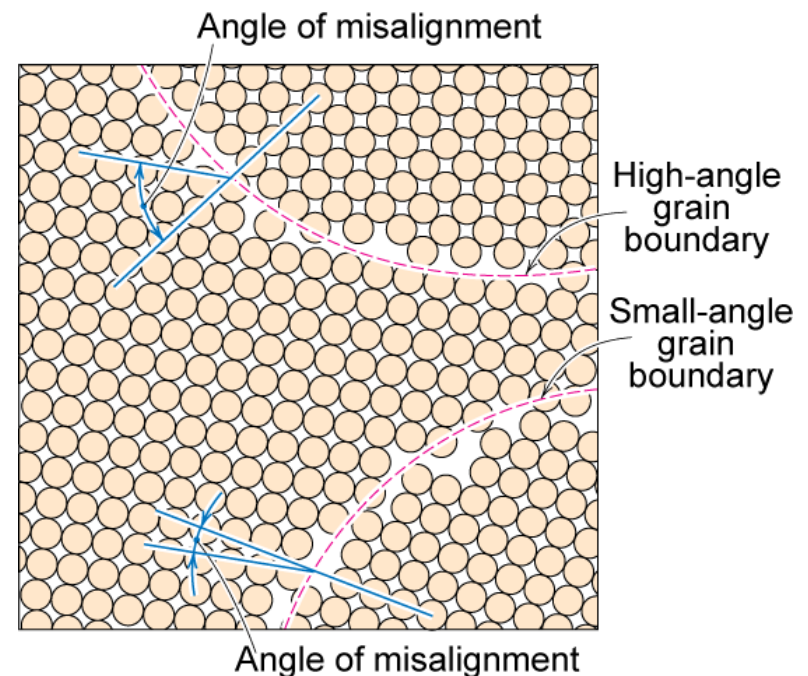
Grain boundaries:

- regions between crystals
- transition from lattice of one region to another



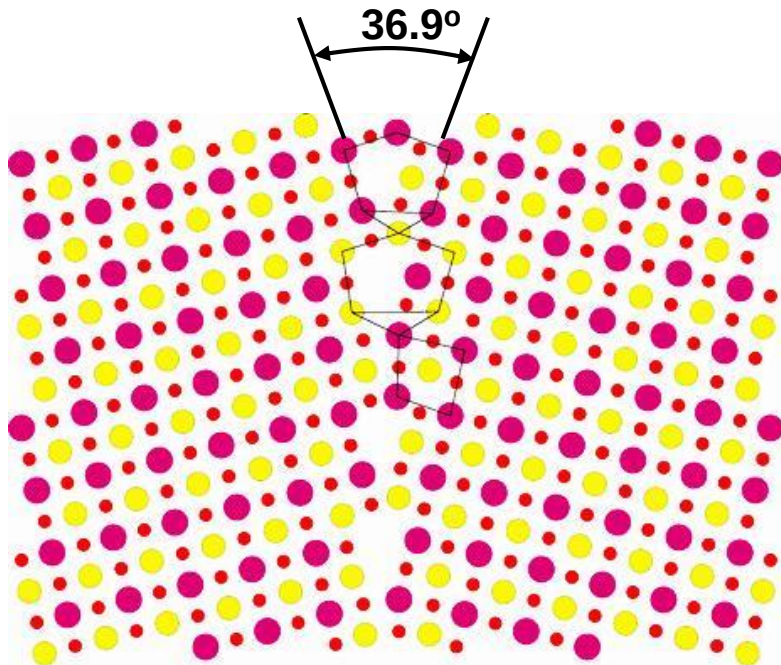
(a) The atoms near the boundaries of the 3 grains do not have an equilibrium spacing or arrangement; slightly disordered.

(b) Grains and grain boundaries in a stainless steel sample. low density in grain boundaries



Planar defects

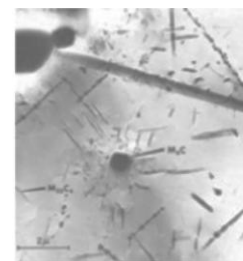
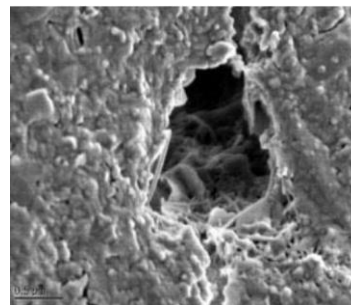
High-angle grain boundaries that have a high density of defects are as strong as the pristine material and unexpectedly are much stronger than those with low-angle boundaries having fewer defects.



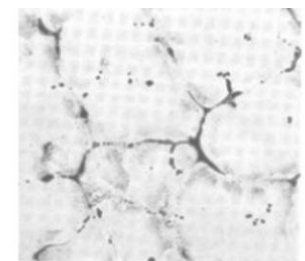
A high-angle ($\theta=36.9^\circ$) grain boundary between two square lattice grains can be represented by a coincident site lattice.

Bulk (volume) defects

- These include **pores, cracks, foreign inclusions, and other phases**. They are normally introduced during processing and fabrication steps.
- **Pores** are small regions where there are no atoms, and can be thought of as clusters of vacancies.
- **Cracks** can greatly affect mechanical properties.
- **Foreign inclusions** can greatly affect electrical, mechanical, optical properties.



Inclusion



Voids/Pores